

CSS

Grid-template areas, opacity, position, pseudo class & element

Grid-area-demo.html

```
<!doctype html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>Grid-area</title>
    <style>
      .container{
        display: grid;
        grid-template-columns: 1fr 2fr 2fr;
        grid-template-rows:1fr 2fr 2fr 1fr;
        height: 96vh;
        grid-template-areas:
          "header header header "
          "side cont1 cont1"
          "side cont2 cont3"
          "footer footer footer";
        width: 100%;
        text-align: center;
        row-gap: 20px;
        column-gap: 10px;
      }
      .box1{
        grid-area:header ;
        background-color: aqua;
      }
      .box2{
        grid-area: side;
        background-color: yellow;
      }
      .box3{
        grid-area:cont1;
        background-color: red;
      }

      .box4{
        grid-area:cont2;
        background-color: green;
      }
      .box5{
        grid-area:cont3;
        background-color: pink;
      }
    </style>
  </head>
  <body>
    <div class="container">
      <div class="box1"></div>
      <div class="box2"></div>
      <div class="box3"></div>
      <div class="box4"></div>
      <div class="box5"></div>
    </div>
  </body>
</html>
```

```

    }
    .box6{
        grid-area:footer;
        background-color: purple;
    }
</style>
</head>
<body>
    <div class="container">
        <div class="box1">
            <h1>Header</h1>
        </div>
        <div class="box2">
            <h1>Side Bar</h1>
        </div>
        <div class="box3">
            <h1>Content 1</h1>
        </div>
        <div class="box4">
            <h1>Content 2</h1>
        </div>
        <div class="box5">
            <h1>content 3</h1>
        </div>
        <div class="box6">
            <h1>Footer</h1>
        </div>
    </div>
</body>
</html>

```

opacityPositon.html

```

<!doctype html>
<html lang="en">
    <head>
        <meta charset="UTF-8" />
        <meta name="viewport" content="width=device-width, initial-scale=1.0" />
        <title>Position</title>

        <style>
            /* h1 {
                opacity: 0.7;
            } */
            /* .container{
                position: relative;
                border: 2px solid black ;
            } */

```

```

div h1 {
  position: absolute;
  background-color: yellow;
  z-index: -100;
  /* top , right, left, bottom */
  top: 50px;
  /* left: 100px; */
  /* width: 100%; */
}
h3 {
  background-color: red;
}

h2 {
  position: fixed;
}
header {
  text-align: center;
  background-color: burlywood;
  padding: 20px;
  position: sticky;
  top: 0;
}
.prod-container{
  position: relative;
}
.prod-container span{
  position: absolute;
  top: 10px;
  left: 10px;
  z-index: 100px;
  color: white;
}
</style>
</head>
<body>
  <div class="prod-container">
    
    <span>100 OFF</span>
  </div>

<!--
  <h1>Opacity</h1>
  <h3>top</h3>

```

Pseudo.html

```
<!doctype html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>Pseudo</title>
    <style>
      h1 {
        color: purple;
        text-transform: capitalize;
      }

      h1:hover {
        color: blue;
      }

      /* ul li:first-child{
        color: red;
      }

      ul li:last-child{
        color: peru ;
      } */

      ul li:nth-child(even) {
        color: aqua;
      }

      a:link {
        color: yellowgreen;
      }

      a:visited {
        color: red;
      }
```

```

a:active {
    color: blue;
}

/* psuedo element -> specific part of the element */
ul li::before {
    content: "😁";
}

ul li::after {
    content: "👁";
}

p::first-letter {
    font-size: 60px;
}

p::first-line{
    color: blue;
}

::selection{
    background-color: orange;
    color: white;
}

::marker{
    font-size: 40px;
    color: rgb(23, 169, 108);
}

</style>
</head>
<body>
<p>
    Lorem ipsum dolor sit amet consectetur adipisicing elit. Sequi qui
    molestiae ea exercitationem similique, dolore corporis delectus quas
    quis
    doloremque hic dolorum ut repudiandae placeat ipsam excepturi, quae
    officiis non?
</p>
<h1>pseudo class</h1>
<a href="#">Google</a>

<ul>
    <li>HTML</li>
    <li>CSS</li>
    <li>Javascript</li>

```

```
<li>React js</li>
<li>HTML</li>
<li>CSS</li>
<li>Javascript</li>
<li>React js</li>
</ul>
</body>
</html>
```

Theory Notes

1. Pseudo-Classes

- **Definition:** Used to style elements based on their state.
- Common pseudo-classes:
 - :hover → when mouse hovers over element.
 - :first-child / :last-child → target first/last child in a parent.
 - :nth-child(n) → target specific child (odd/even or index).
 - :link, :visited, :active → link states.

2. Pseudo-Elements

- **Definition:** Style specific parts of an element.
- Common pseudo-elements:
 - ::before / ::after → insert content before/after element.
 - ::first-letter → style the first letter.
 - ::first-line → style the first line.
 - ::selection → style text when selected.
 - ::marker → style list item markers.

3. Positioning

- **Static** → default, follows normal flow.
- **Relative** → positioned relative to itself.
- **Absolute** → positioned relative to nearest positioned ancestor.
- **Fixed** → positioned relative to viewport, stays fixed when scrolling.
- **Sticky** → toggles between relative and fixed depending on scroll.
- **z-index** → controls stacking order.

4. Grid-Area

- **Definition:** CSS Grid allows naming areas and placing items accordingly.
- Properties:
 - grid-template-areas → defines layout template.
 - grid-area → assigns element to a named area.
- Useful for complex page layouts (header, sidebar, content, footer).

Practice Tasks

Short Answer

1. What is the difference between pseudo-class and pseudo-element?
2. How does position: absolute differ from position: fixed?
3. Why is z-index important in positioning?
4. How does grid-template-areas simplify layout design?

Coding Exercises

1. Style the first child of a list in red and the last child in green.
2. Add an emoji before and after each list item using ::before and ::after.
3. Create a sticky navigation bar that stays at the top when scrolling.
4. Build a product card with an absolute-positioned “SALE” badge.
5. Use grid-template-areas to design a layout with header, sidebar, content, and footer.
6. Style the first letter of a paragraph to be larger and bold.
7. Use ::selection to change background color when text is highlighted.